

Cauchy-Schwarz inequality in $\mathbb{R}^n$. Let

$$\mathbf{a} = (a_1, \dots, a_n), \quad \mathbf{b} = (b_1, \dots, b_n) \in \mathbb{R}^n.$$

The standard inner product and Euclidean norm are

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^n a_i b_i, \quad \|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle} = \left(\sum_{i=1}^n a_i^2 \right)^{1/2}.$$

For any real t consider the non-negative quadratic form

$$f(t) = \|\mathbf{a} - t\mathbf{b}\|^2 = \langle \mathbf{a} - t\mathbf{b}, \mathbf{a} - t\mathbf{b} \rangle = \sum_{i=1}^n (a_i - tb_i)^2 \geq 0.$$

Expanding the sum gives

$$\begin{aligned} f(t) &= \sum_{i=1}^n (a_i^2 - 2ta_i b_i + t^2 b_i^2) \\ &= \left(\sum_{i=1}^n b_i^2 \right) t^2 - 2 \left(\sum_{i=1}^n a_i b_i \right) t + \sum_{i=1}^n a_i^2. \end{aligned}$$

Set

$$A := \sum_{i=1}^n b_i^2, \quad B := -2 \sum_{i=1}^n a_i b_i, \quad C := \sum_{i=1}^n a_i^2,$$

so that $f(t) = At^2 + Bt + C$.

Since $f(t) \geq 0$ for all $t \in \mathbb{R}$, the quadratic $At^2 + Bt + C$ cannot have two distinct real roots; consequently its discriminant is non-positive:

$$B^2 - 4AC \leq 0.$$

(If $A = 0$ then $\mathbf{b} = 0$, whence $B = 0$ and the inequality is trivial.)

Substituting A, B, C yields

$$\left(-2 \sum_{i=1}^n a_i b_i \right)^2 - 4 \left(\sum_{i=1}^n b_i^2 \right) \left(\sum_{i=1}^n a_i^2 \right) \leq 0.$$

Simplifying,

$$4 \left(\sum_{i=1}^n a_i b_i \right)^2 \leq 4 \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

Dividing by $4 > 0$ we obtain the ****Cauchy-Schwarz inequality****

$$\boxed{\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right)}.$$

Equality case. Equality holds iff the discriminant vanishes, i.e. $B^2 = 4AC$. In that situation the quadratic $f(t)$ has a double root at

$$t_0 = \frac{\sum_{i=1}^n a_i b_i}{\sum_{i=1}^n b_i^2} \quad \left(\text{provided } \sum_{i=1}^n b_i^2 \neq 0 \right).$$

Then $f(t_0) = 0$, so $\mathbf{a} = t_0 \mathbf{b}$. Hence there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i , i.e. the two vectors are linearly dependent. If $\mathbf{b} = \mathbf{0}$ then both sides of the inequality are zero, and equality holds trivially; this is consistent with the statement that equality occurs precisely when the vectors are linearly dependent (the zero vector being linearly dependent with any vector).

Thus, for any real families $(a_i)_{i=1}^n$ and $(b_i)_{i=1}^n$,

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right),$$

with equality exactly when $\mathbf{a}$ and $\mathbf{b}$ are linearly dependent. This is the Cauchy–Schwarz inequality for real sequences. $\square$